

An Inquiry-Based Introduction to Proofs

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Preface

This is a course in mathematical proof. It is for math majors, typically sophomores. Its only prerequisite is high school mathematics.

Approach. This course is Inquiry-Based (sometimes called the Discovery Method or Moore Method). Students get a sequence of exercises along with definitions and a few remarks. They attempt these outside of class, and then in class propose solutions. The instructor only lightly guides and the students pledge not to use outside sources so they together talk through misunderstandings, sometimes stumble in the dark, and sometimes have beautiful flashes of insight. Most important is that each person must grapple directly with the mathematics. For these students, with this material, this is the best way to develop mathematical maturity. Besides, it is a great deal of fun.

Topics. We cover elementary number theory, sets, functions, and relations.

We start with number theory for the same reason that the baseball team's annual practice starts by tossing the ball and not by reading the rule book. Math majors take readily to proving things about divisibility and primes, whereas weeks of preliminary material is less of a lure.

But the background is good stuff also and students get on board quickly. The second and third chapters does the other material, keeping the intellectual habits that we established at the start.

Exercises. Where possible, nearby exercises have about the same difficulty. This level gradually rises.

Some exercises have multiple items; these come in two types. If the items are labeled A, B, etc., then each one is hard enough to be a separate assignment. If the labels are (i), (ii), etc., then they together make a single assignment. I have students put proposed answers on the board for discussion and if the items are labeled alphabetically then I ask a different student to do each one while for the others I ask a single student to do them all.

License. This book is Free; see <http://joshua.smcvt.edu/proofs>. In particular, instructors are free to copy the text and hand out those copies.

Acknowledgments. The material is standard but I must recognize my debt to the wonderful presentations of J Jensen-Vallin and D Velleman. I am also glad to express my debt to the Saint Michael's students in Math 240, who have taught me a great deal.

The most important thing [is that] proving things in math [i]s a skill like any other that you get good at through practice. —Cathy O'Neil

It's a kind of art that may change lives. —Peter Schjeldahl

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For instructors

These notes can be used in a variety of ways but perhaps the most helpful approach would be if I say how I use them. I cover all of the first three chapters, averaging four assignments per session. The fourth chapter is suitable as a lecture on the final day or two, or perhaps as extra credit.

I will run through a typical class session. As background, on the first day I explain that in this course we are developing each persons's ability to work independently. Students pledge that outside of class they will not work together and will not use any resources such as other books or the Internet. Several times during the semester I refer to the pledge, to keep folks on track and also to remind them that others are doing the same.

Consequently, on each day each student arrives having worked on each exercise, on their own. Usually they have four exercises. I shuffle cards to pick students to put proposed solutions on the board. (They negotiate among themselves who will do which problem, a student picked in the prior class will not be picked in this one, and each student is twice allowed to take a pass).

With that, the work starts. The class as a whole discusses those answers sentence by sentence, and sometimes word by word. These discussions are filled with misconceptions to work through and ideas that folks eventually come to see are not fruitful, as well as good ideas that, perhaps, initially have trouble getting heard. But the discussion eventually comes to an end and usually that end is correct.

I do my best to speak as little as I can, and not even to nod or frown. Sometimes I must guide, as with speaking up after the class has decided that the proof is OK to say, "Exhibiting the $n = 3$ and $n = 5$ cases is not enough to prove the result for all odd numbers."

The advantage of this format over a lecture is twofold. First, students own it. Second, through its evolution students come not only to see what is right, but also to understand why what's wrong is wrong.

The discussion often takes the entire period. But sometimes we have time

for another problem; by experience I know that some (such as proving that there are infinitely many primes) can give folks trouble and I try arrange the assignments so that we do these in class, where I can do a little extra guiding.

With that, class ends with a new set of problems to work on at home, and perhaps once a week with the announcement that one of them is to be handed in at the start of the next class, to be graded.

I will close with a few comments. First, in the first few classes while folks are putting their answers on the board, I spend fifteen minutes on slides that cover elementary logic. These provide a vocabulary and sharpen the discussions, particularly about fine points such as vacuous implication.

Second, we run this class for sophomores who have had Calculus III and often Linear Algebra, so we can expect that students have aptitude. We have fewer than twenty students because if the class is too big then people can hide. And besides, with more than that the grading burden is prohibitive.

Finally, about grading. I grade on four things: class contribution, weekly hand-in problems, an in-class midterm, and an in-class final. I give credit both for mathematical correctness and for style. (The style that I ask for is simply full sentences, not beginning with a symbol, in paragraphs. And, over the semester I move students to formatting hand-in problems with $\text{\LaTeX}$.) I weigh the four grades equally but students agree that discussion of the problems is the core experience. Success in the class flows from that.

More material. As mentioned in the Preface, this book is Free. You are free to copy the text and hand out those copies. For more information see <http://joshua.smcvt.edu/proofs>.

That page has other material that you may find useful, including the classroom slides that I use and a link to the book's $\text{\LaTeX}$ source, so you can fine tune the material to suit your class.

Also there are other versions of the text that you may prefer. In particular, one of them is compact—it fits on seven double-sided sheets that you can just hand out on the first day as the text. That is what I do.

Contact, and updates. I am always glad to get reports about this book, either a description of your experience, or suggestions, or bug reports. I save these and periodically revise. Please email me at the address on the book's page, <http://joshua.smcvt.edu/proofs>.

For students

In this course you will write proofs. At the start you will be asked to prove things that you already know, so you can get used to the process of proving in a context where you are not also struggling with the material. Later you will prove things that are new to you.

In any event, you need to know on what you can base the proofs, that is, what you can assume. For one thing, you can always use a prior result. You can also use the rules of high school algebra such as associativity of addition $x + (y + z) = (x + y) + z$, or distributivity of multiplication over addition $x \cdot (y + z) = xy + xz$, or that a positive times a positive equals a positive. Finally, you can use elementary logic, such as that a statement like “ P and Q ” is true if and only if both halves are true. Perhaps your instructor will go over some of the logic but in any event the proofs that you will start with require only logic that is natural for people with a mathematical turn of mind.

Chapter 1

Numbers

We begin with results about the integers $\mathbb{Z} = \{\dots -2, -1, 0, 1, 2, \dots\}$. In this chapter, “number” means integer. Some statements refer to the natural numbers $\mathbb{N} = \{0, 1, 2, \dots\}$ or the positive integers $\mathbb{Z}^+ = \{1, 2, \dots\}$.

1.1 Divisibility

Definition 1. For two integers d, n we say that d divides n if there is an integer k such that $d \cdot k = n$. Here, d is the divisor, n is the dividend, and k is the quotient. (Alternative wordings are: d is a factor of n , d goes evenly into n , n is divisible by d , or n is a multiple of d .) We denote the relationship as $d \mid n$ if d is a divisor of n or as $d \nmid n$ if it is not.

Definition 2. A number is even if it is divisible by 2, otherwise it is odd. (We may instead say that the parity is even or odd.)

The notation $d \mid n$ signifies a relationship between two integers. Note that it is different than the fraction d/n , which is a rational number. We can sensibly ask “Does 2 divide 5?” but “Does $2/5$?” is not sensible.

Problem 3. (Interaction with sign) Each of these is a statement about integers. Prove each.

- A. If a number is even then its negative is even. If a number is odd then its negative is odd.
- B. If $d \mid a$ then $-d \mid a$ and $d \mid -a$. In addition, $d \mid |a|$ (recall that the absolute value of a number $|a|$ is a if $a \geq 0$ and is $-a$ if $a < 0$).

Problem 4. (Interaction of parity and addition) Prove or disprove.

- A. The sum of two evens is even. The difference of two evens is even.

- B. The sum of two odds is odd. The difference of two odds is odd.*
- C. Where $a, b \in \mathbb{Z}$, the number $a + b$ is even if and only if $a - b$ is even.*
- D. Generalize the first item to be a statement about sums of multiples of some $d \in \mathbb{Z}$. Prove it.*

Problem 5. *(Interaction of parity and multiplication) Prove or disprove.*

- A. The product of two evens is even. Formulate and prove an analogous statement that applies to any integer.*
- B. The quotient of two evens, if it is an integer, is even.*

Problem 6. *(Divisibility Properties) Prove each. Assume that all the numbers are integers.*

- A. (Reflexivity) Every number divides itself.*
- B. Every number divides 0 while the only number that 0 divides is itself.*
- C. (Transitivity) If $d \mid n$ and $n \mid m$ then $d \mid m$. That is, if n divides m then so do n 's divisors.*
- D. (Cancellation) Where $d, n \in \mathbb{Z}$, if there is a nonzero integer a such that $ad \mid an$ then $d \mid n$. And, if $d \mid n$ then $ad \mid an$ for all $a \in \mathbb{Z}$.*
- E. (Comparison) For $d, n \in \mathbb{Z}^+$, if n is a multiple of d then $n \geq d$.*
- F. Every number is divisible by 1 and -1 . The only numbers that divide 1 are 1 and -1 .*
- G. The largest divisor of a is $|a|$, for $a \in \mathbb{Z}$ with $a \neq 0$.*
- H. Every nonzero integer has only finitely many divisors.*

Problem 7. *What conclusion can you make if $a \mid b$ and $b \mid a$?*

Problem 8. *Suppose that $a, b, c \in \mathbb{Z}$.*

- A. Prove that if $a \mid b$ then $a \mid bc$ for all integers c .*
- B. Prove that if $a \mid b$ and $a \mid c$ then a divides the sum $b + c$ and difference $b - c$.*
- C. (Linearity) Prove that if $a \mid b$ and $a \mid c$ then a divides $m \cdot b + n \cdot c$ for any $m, n \in \mathbb{Z}$.*

1.2 Interlude: induction

Results in the prior section need only proof techniques that come naturally to people with mathematical aptitude. However some results to follow require a technique that is less natural, mathematical induction. This section introduces induction. We will start with exercises about summations. (However, note that induction is not about summation; they just make good exercises.)

For example, in playing with numbers many people have noticed that the odd natural numbers sum to perfect squares: $1 + 3 = 4$, $1 + 3 + 5 = 9$, $1 + 3 + 5 + 7 = 16$, etc. We will prove the statement, “The sum $1 + 3 + 5 + \cdots + (2n + 1)$ equals $(n + 1)^2$.”

That statement has a natural number variable n that is free, meaning that setting n to be 0, or 1, etc., gives a family of statements: $S(0)$, or $S(1)$, etc. For instance, the statement $S(1)$ asserts that $1 + 3$ equals 2^2 . Our induction proofs will all involve statements with one free natural number variable.

These proofs have two steps. For the *base step* we will show that the statement holds for some initial number $i \in \mathbb{N}$ (sometimes there is a finite list of initial numbers). The *inductive step* is more subtle; we show that the following implication holds.

If the statement holds from the initial number up to and including $n = k$
then the statement holds also in the next case, where $n = k + 1$.

(*)

The *Principle of Mathematical Induction* is that doing both steps proves that the statement is true for all natural numbers greater than or equal to the initial number i .

For the example statement about odd numbers and squares, the intuition behind the principle is first that the base step directly verifies the statement for the initial number 0. Next, because we have shown that the implication (*) holds in all cases, applied to the $k = 0$ case it gives that the statement holds also for the number 1. That is, (*) with $k = 0$ says that $S(0)$ implies $S(1)$, and because we have verified the assertion $S(0)$, we conclude that $S(1)$ holds. Continuing on, (*) with $k = 1$ says that $S(0)$ and $S(1)$ together imply $S(2)$, so we know that $S(2)$ holds. In this way, induction bootstraps to all numbers. (We sometimes instead use induction to show that a statement is true for all numbers in a finite interval.)

Here is an induction argument for the example statement.

Proof. We show that $1 + 3 + \cdots + (2n + 1) = (n + 1)^2$ by induction. For the $n = 0$ base step note that the sum on the left has a single term, 1, which equals the value on the right, 1^2 .

For the inductive step assume that the formula is true for $n = 0, n = 1, \dots, n = k$, and consider the $n = k + 1$ case. The sum is $1 + 3 + \dots + (2k + 1) + (2(k + 1) + 1) = 1 + 3 + \dots + (2k + 1) + (2k + 3)$. By the inductive hypothesis the statement is true in the $n = k$ case so we can substitute $1 + 3 + \dots + (2k + 1) + (2k + 3) = (k + 1)^2 + (2k + 3) = (k^2 + 2k + 1) + (2k + 3) = (k + 2)^2$. This is the required expression for the $n = k + 1$ case. ■

Problem 9. *Prove by induction.*

- A. $0 + 1 + 2 + \dots + n = n(n + 1)/2$
- B. $0 + 1 + 4 + 9 + \dots + n^2 = n(n + 1)(2n + 1)/6$
- C. $1 + 2 + 4 + 8 + \dots + 2^n = 2^{n+1} - 1$

Problem 10. *Prove each by induction. Suppose that $a, b, r \in \mathbb{R}$ and that $r \neq 1$.*

- A. (Geometric Series) $1 + r + r^2 + \dots + r^n = (r^{n+1} - 1)/(r - 1)$
- B. (Arithmetic Series) $b + (a + b) + (2a + b) + \dots + (na + b) = (n(n + 1)/2) \cdot a + (n + 1) \cdot b$

Problem 11. *Prove by induction that $n < 2^n$ for all $n \in \mathbb{N}$.*

Problem 12. *Prove each by induction.*

- A. *For all $n \in \mathbb{N}$, the number $n^2 + n$ is even.*
- B. *For all $n \geq 2$ the number $n^3 - n$ is divisible by 6. Hint: use $n = 2$ for the base step.*
- C. *If $n \in \mathbb{Z}^+$ then $(1 + 1/1) \cdot (1 + 1/2) \cdot \dots \cdot (1 + 1/n) = n + 1$.*

Problem 13. *Prove that n -term sums of reals commute: $a_0 + a_1 + \dots + a_{n-1} = a_{n-1} + \dots + a_0$ for all $n \geq 2$, starting from the assumption that sum of two terms commutes.*

While many induction arguments use only the $n = k$ part of the inductive hypothesis, some break from that pattern.

Problem 14. *The game of Nim starts with two piles, each with n chips. The two players take turns picking a pile, and taking from it a nonzero number of chips. The player taking the final chip wins. Prove that by using this strategy the second player always wins: whatever number of chips the first player takes from one pile, the second player takes the same number from the other pile.*

Problem 15. The Fibonacci number sequence $0, 1, 1, 2, 3, 5, 8, 13, \dots$ satisfies that each succeeding number is the sum of the prior two $f_{n+1} = f_{n-1} + f_n$, with $f_0 = 0$ and $f_1 = 1$. The following argument purports to show that all Fibonacci numbers are even. Where does it go wrong? “The base step is clear since 0 is even. For the inductive step, assume the statement is true for all cases up to and including $n = k$. By definition the next case f_{k+1} is the sum of the two prior numbers, which by the inductive hypothesis are both even. Thus their sum is even.”

Definition 16. The Least Number Principle, or Well-Ordering Principle, is that any nonempty subset of the natural numbers has a least element.

Problem 17. Prove each.

- A. The Principle of Induction implies the Least Number Principle. Hint: use induction on $n \in \mathbb{N}$ to show that if a set of natural numbers contains n then it has a least element.
- B. The Least Number Principle implies the Principle of Induction. Hint: start with a set that contains 0 and such that if it contains $0, \dots, k$ then it contains $k + 1$, and show this set equals $\mathbb{N}$.

1.3 Division

Problem 18. (Division Theorem, or Division Algorithm) For any integers a, b with $b > 0$ there are unique integers q, r such that $a = bq + r$ and $0 \leq r < b$. Here, a is the dividend and b is the divisor, while q is the quotient and r is the remainder. We prove this statement in three stages.

- A. Show that q and r are unique, assuming that they exist. Hint: one way to proceed is to suppose that $a = bq_0 + r_0 = bq_1 + r_1$ with $0 \leq r_0, r_1 < b$ and then deduce that $q_0 = q_1$ and $r_0 = r_1$.
- B. Verify that there exists such a q, r pair when $a = 0$. Show that if the statement holds when $a > 0$ then it holds when $a < 0$.
- C. Prove the statement for $a > 0$. Hint: show that the set $\{a - bq \mid q \in \mathbb{Z}\}$ has nonnegative elements, apply the Least Number Principle to get a smallest one r , and verify that this has the properties required of a remainder.

Observe that $r = 0$ if and only if $b \mid a$.

Definition 19. Where $m > 0$, the remainder when a is divided by m is the modulus, $a \bmod m$. Two numbers a, b are congruent modulo m , written $a \equiv b \pmod{m}$, if they leave the same remainder when divided by m , that is, if they have the same modulus with respect to m .

Problem 20. Find $5 \bmod 3$ and $-5 \bmod 3$.

Problem 21. Prove or disprove.

i. $a \bmod b = b \bmod a$

ii. $a \bmod b = -a \bmod b$

Problem 22. Prove that $a \equiv b \pmod{m}$ if and only if $m \mid (a - b)$, that is, if and only if a and b differ by a multiple of m , where $m > 0$.

Problem 23. Let a, b, c, d, m be integers with $m > 0$, and $a \equiv b \pmod{m}$, and $c \equiv d \pmod{m}$. Prove each.

A. $a + c \equiv b + d \pmod{m}$

B. $ac \equiv bd \pmod{m}$

C. $a^n \equiv b^n \pmod{m}$ for all powers $n \in \mathbb{Z}^+$

Definition 24. For any real number x , its floor $\lfloor x \rfloor$ is the greatest integer less than or equal to x .

Problem 25. Prove each, where $a, b, m \in \mathbb{Z}$ and $b > 0$.

A. The quotient when a is divided by b is $\lfloor a/b \rfloor$. Thus $a = b \cdot \lfloor a/b \rfloor + a \bmod b$.

B. $b \cdot (a \bmod m) = (ba) \bmod (bm)$

1.4 Common divisors and common multiples

Definition 26. A common divisor of $a, b \in \mathbb{Z}$ is an integer that divides both. Where at least one of them is nonzero, their greatest common divisor, $\gcd(a, b)$, is the largest of their common divisors.

Problem 27. Prove.

A. (Existence) For any two integers a, b that are not both zero, $\gcd(a, b)$ exists and is positive.

B. (Commutativity) $\gcd(a, b) = \gcd(b, a)$

C. If d is a common divisor of a and b then so is $-d$. Thus common divisors are restricted to the interval from $-\gcd(a, b)$ to $\gcd(a, b)$, inclusive.

D. $\gcd(a, b) = \gcd(|a|, |b|)$

E. If both numbers are nonzero then $0 < \gcd(a, b) \leq \min(|a|, |b|)$. If either number is zero then the greatest common divisor is the absolute value of the other.

Definition 28. Two integers are relatively prime or coprime, sometimes denoted $a \perp b$, if their greatest common divisor is 1.

Definition 29. The least common multiple of two positive integers $\text{lcm}(a, b)$ is the smallest positive integer that is a multiple of each.

Problem 30. Prove each.

i. (Existence) Any two positive integers have a least common multiple.

ii. (Commutativity) $\text{lcm}(a, b) = \text{lcm}(b, a)$

Problem 31. (Euclid's Algorithm) Prove that if $a = bq + r$ then $\gcd(a, b) = \gcd(b, r)$.

The algorithm associated with this result quickly finds the greatest common divisor for any $a, b \in \mathbb{N}$. For instance, to find $\gcd(803, 154)$ divide the larger by the smaller $803 = 154 \cdot 5 + 33$ and then the prior result gives that $\gcd(803, 154) = \gcd(154, 33)$. Iterate: since $154 = 33 \cdot 4 + 22$, we have that $\gcd(154, 33) = \gcd(33, 22)$. Continuing gives $33 = 22 \cdot 1 + 11$ and $\gcd(33, 22) = \gcd(22, 11)$, and the last step is that $22 = 11 \cdot 2 + 0$ shows that $\gcd(22, 11) = 11$. The zero remainder signals that we are done, and $\gcd(803, 154) = 11$.

Reversing this calculation is also fruitful. Start by rewriting $33 = 22 \cdot 1 + 11$ to put the greatest common divisor on the left, $11 = 1 \cdot 33 - 1 \cdot 22$. Next, rewrite the next equation $154 = 33 \cdot 4 + 22$ to isolate its remainder and substitute: $11 = 1 \cdot 33 - 1 \cdot (154 - 4 \cdot 33) = -1 \cdot 154 + 5 \cdot 33$. Similarly using $803 = 154 \cdot 5 + 33$ gives $11 = -1 \cdot 154 + 5 \cdot (803 - 5 \cdot 154) = 5 \cdot 803 - 26 \cdot 154$. This expresses the greatest common divisor 11 as a combination of the initial numbers 803 and 154.

Definition 32. A number c is a linear combination of two others a and b if it has the form $c = a \cdot m + b \cdot n$ for some $m, n \in \mathbb{Z}$.

Problem 33. Use Euclid's Algorithm to find the greatest common divisor, and then reverse that to express the greatest common divisor as a linear combination of the two.

i. 123, 54

ii. 48, 732

Problem 34. *Prove.*

- A. *The greatest common divisors of two numbers is a linear combination of the two.*
- B. *(Bézout's Lemma) The greatest common divisor of two numbers is the smallest positive number that is a linear combination of the two. Hint: consider the set of all combinations.*

Problem 35. *You are given three buckets. The first two are marked 6 liters and 11 liters while the last one, which is quite large, is unmarked. Taking water from a nearby pond, use those buckets to end with 8 liters in the unmarked one.*

Problem 36. *Prove each, for $a, b, c \in \mathbb{Z}$ and $m \in \mathbb{N}$.*

- A. $\gcd(ma, mb) = m \cdot \gcd(a, b)$
- B. *If at least one of $a, b \in \mathbb{Z}$ is not zero and d is a common divisor then $\gcd(a/d, b/d) = \gcd(a, b)/d$. Thus, $a/\gcd(a, b)$ and $b/\gcd(a, b)$ are relatively prime.*
- C. *(Euclid's Lemma) If a and b are relatively prime then $a \mid bc$ implies that $a \mid c$.*

1.5 Primes

Definition 37. *A natural number greater than 1 is prime if its only positive divisors are 1 and itself. A natural number greater than 1 that is not prime is composite.*

Problem 38. *Verify each.*

- i. *There are 25 primes less than 100.*
- ii. *Below 50 there are 6 pairs of twin primes, primes separated by 2.*
- iii. *The numbers $2^{2^0} + 1, \dots, 2^{2^4} + 1$ are prime.*

Problem 39. *Prove.*

- A. *A number n is composite if and only if it can be decomposed into the product of two factors $n = a \cdot b$ such that $1 < a, b < n$ (the two might be equal).*

- B. Every number greater than 1 has a prime divisor.
- C. Every composite number n has a prime divisor p with $p \leq \sqrt{n}$. This inequality cannot be made strict.

Problem 40. (Euclid's Theorem) There are infinitely many primes.

Problem 41. Let p be prime. Prove each.

- A. If $p \mid ab$ then either $p \mid a$ or $p \mid b$.
- B. If $p \mid a_0 \cdot a_1 \cdots a_{n-1}$ for $n \geq 2$ then p divides at least one a_i .

Problem 42. (Fundamental Theorem of Arithmetic) Any number $n > 1$ can be factored into primes, $n = p_0^{e_0} p_1^{e_1} \cdots p_s^{e_s}$. What's more, if the primes are distinct then this expression is unique: for each prime p_i , any two prime factorizations of n have the same power e_i of p_i . We prove this in two parts.

- A. Prove that any $n > 1$ can be written as a product of primes.
- B. Prove that the factorization is unique. Hint: show that where $n = p_0 \cdots p_i$ is a product of (possibly not distinct) primes, if $n = q_0 \cdots q_j$ is also a product of primes then the primes are the same, with the same multiplicities, possibly rearranged; you can use induction on i .

Remark: this result is why we do not include 1 among the primes. Including 1 would require us to change the clause about uniqueness, since we can always multiply by additional 1's.

Problem 43. True or false?

- i. $5 \cdot 7 \cdot 19 = 3 \cdot 11 \cdot 17$
- ii. $1357 \cdot 4183 = 1081 \cdot 5251$

Problem 44. Let $a = p_0^{e_0} \cdots p_n^{e_n}$ and $b = p_0^{f_0} \cdots p_n^{f_n}$ express each number as a product of distinct primes; to use the same primes $p_0, \dots, p_n$ in both we allow here that some exponents are zero. Prove that in the prime factorization of $\gcd(a, b)$ the exponent of p_i is $\min(\{e_i, f_i\})$. (Much the same proof shows that in the prime factorization of $\text{lcm}(a, b)$ the exponent of p_i is $\max(\{e_i, f_i\})$. Together these show that $\gcd(a, b) \cdot \text{lcm}(a, b) = ab$.)

Problem 45. (Existence of Irrational Numbers)

- A. Prove that in the prime factorization of a square, each prime is raised to an even power.
- B. Conclude that $\sqrt{2}$ is irrational.

Chapter 2

Sets

A *set* is a collection that is definite, so that every thing either definitely is contained in the collection or definitely is not. An object x that belongs to a set A is an *element* or *member* of that set, denoted $x \in A$. Denote that x is not an element with $x \notin A$. Sets are equal if and only if they have the same elements. (Read ‘ $\in$ ’ as “is an element of” rather than “in” to avoid confusion between this and the subset relation defined below.)

We usually describe a set either by listing its elements or by stating a criteria for membership. Thus we can write the set of primes less than ten as $\{2, 3, 5, 7\}$ or as $\{p \in \mathbb{N} \mid p \text{ is prime and } p < 10\}$ (read the vertical bar as “such that”; some authors instead use a colon, ‘:’).

Problem 46. *Decide if each is true and justify your decision.*

- i. $\{1, 3, 5\} = \{5, 3, 1\}$
- ii. $\{2, 4, 6\} = \{2, 4, 6, 4\}$
- iii. $\{1, 3\} = \{n \in \mathbb{N} \mid n < 5\}$
- iv. $0 \in \{1, 2, \{0\}\}$
- v. $4 \in \{n \in \mathbb{N} \mid n^2 < 50\}$

Definition 47. *The set B is a subset of the set A if every element of B is an element of A , that is, provided that $x \in B$ implies that $x \in A$. We write $B \subseteq A$.*

Definition 48. *The set without any elements is the empty set, denoted $\emptyset$.*

Problem 49. *Decide whether or not each is true. Justify your answer.*

- i. $\{1, 3, 5\} \subseteq \{1, 3, 5, 7, 9\}$
- ii. $\{1, 3, 5\} \in \{1, 3, 5, 7, 9\}$

iii. $\{1, 3, 5\} \subseteq \{n \in \mathbb{N} \mid n \text{ is prime}\}$

iv. $\emptyset \subseteq \{1, 2, 3, 4\}$

v. $\{2\} \in \{1, \{2\}, 3\}$

vi. $\{2\} \subseteq \{1, \{2\}, 3\}$

Problem 50. *Prove.*

A. For all sets A , both $A \subseteq A$ and $\emptyset \subseteq A$

B. The empty set is unique: if the set A is empty and the set B is empty then $A = B$.

Problem 51. *Prove, for all sets A , B , and C .*

A. (Mutual Inclusion) If $A \subseteq B$ and $B \subseteq A$ then $A = B$.

B. (Transitivity) If $A \subseteq B$ and $B \subseteq C$ then $A \subseteq C$.

Mutual inclusion is the most common way to show that two sets are equal.

Problem 52. *For each, give an example of three sets satisfying the conditions, or show that no example is possible.*

i. $A \subseteq B$, $B \not\subseteq C$, $A \subseteq C$

ii. $A \not\subseteq B$, $B \not\subseteq C$, $A \subseteq C$

iii. $A \not\subseteq B$, $B \subseteq C$, $A \subseteq C$

Usually mathematical statements are in a context of some *universe*, denoted Ω . For instance, in the first chapter the universe is the integers, $\Omega = \mathbb{Z}$. There, if we say that we are considering the set of things less than 100 then we are considering the set of integers less than 100. Another example is that in first semester calculus the universe is the set of real numbers, $\Omega = \mathbb{R}$.

Problem 53. (Russell's Paradox) *The definition that we gave allows sets to contain anything. This turns out to be naive. For, if sets can contain anything then we naturally think of the set that contains everything, all sets. Note that it contains itself as an element. In this way we are led to the set of all sets that don't contain themselves $D = \{S \mid S \notin S\}$.*

A. Show that assuming D is an element of itself leads to a contradiction.

B. Show that assuming D is not an element of itself also leads to a contradiction.

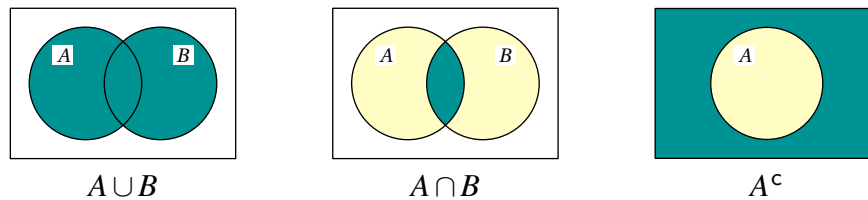
2.1 Operations

Definition 54. The complement of a set A , denoted A^c , is the set of things that are not elements of A . Some authors denote the complement with a bar, $\bar{A}$.

Remark: working inside of a universe makes the complement sensible. For instance, in a number theory discussion where $\Omega = \mathbb{Z}$, if we consider the set of things less than 100 then we can take the complement and the result is another subset of Ω , so we are still in number theory.

Definition 55. Let A and B be sets. Their union is the collection of elements from either set, $A \cup B = \{x \mid x \in A \text{ or } x \in B\}$. Their intersection is the collection of elements from both sets, $A \cap B = \{x \mid x \in A \text{ and } x \in B\}$.

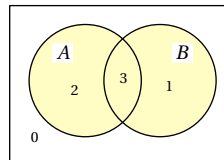
Picture set operations with Venn diagrams.



In each diagram the region inside the rectangle depicts the universe and the region inside a circle depicts a set. On the left the darker color shows the union as containing all of the two sets, the middle shows the intersection containing only the region common to both, and on the right the complement is all but the set A .

Problem 56. Another tool illustrating set relationships is this table. It describes two sets in the universe $\Omega = \{0, 1, 2, 3\}$. (It uses 0 and 1 in place of F and T so that the right side of each row is the binary representation of its number.)

Number	$x \in A$	$x \in B$
0	0	0
1	0	1
2	1	0
3	1	1



The table's top row says that $0 \notin A$ and $0 \notin B$, while the second row says that $1 \notin A$ but $1 \in B$. For each of the following simple results about set operations, apply the statement to these example sets. Also, pick one statement and prove it.

- i. The complement of the complement is the original set, $(A^c)^c = A$.
- ii. $A \cap \emptyset = \emptyset$ and $A \cup \emptyset = A$
- iii. (Idempotence) $A \cap A = A$ and $A \cup A = A$
- iv. $A \cap B \subseteq A \subseteq A \cup B$
- v. (Commutativity) $A \cap B = B \cap A$ and $A \cup B = B \cup A$
- vi. (Associativity) $(A \cap B) \cap C = A \cap (B \cap C)$ and $(A \cup B) \cup C = A \cup (B \cup C)$
(extend the table to add a third set, C).

Problem 57. Prove that the following statements are equivalent: (i) $A \subseteq B$, (ii) $A \cup B = B$, and (iii) $A \cap B = A$. (Hint: one approach is to show that (i) implies (ii), that (ii) implies (iii), and that (iii) implies (i).)

Definition 58. The difference of two sets is $A - B = \{x \in A \mid x \notin B\}$. The symmetric difference is $A \triangle B = (A - B) \cup (B - A)$.

Remark: if $A \subseteq X$ then $X - A$ is the same as A^c where X is the universe, $\Omega = X$.

Problem 59. Prove or disprove.

- A. For any two sets, $A - B = A \cap B^c$ (and thus $A - B \subseteq A$).
- B. For all pairs of sets, $A - B = B - A$.
- C. For all pairs of sets, $A \triangle B = B \triangle A$.

Problem 60. Prove, for all sets A , B , and C .

- A. (De Morgan's Laws) $(A \cap B)^c = A^c \cup B^c$ and $(A \cup B)^c = A^c \cap B^c$
- B. (Distributivity) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ and $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

Definition 61. Two sets are disjoint if their intersection is empty.

Problem 62. Find three sets A , B , and C , such that $A \cap B \cap C$ is empty but the sets are not pairwise disjoint, that is, $A \cap B$, $A \cap C$, and $B \cap C$ are all nonempty.

Definition 63. For a finite set A , the order $|A|$ is the number of elements.

Problem 64. For finite sets, if $A \subseteq B$ then $|A| \leq |B|$.

Definition 65. For a set A , the power set $\mathcal{P}(A)$ is the set of all subsets of A .

Problem 66. List the elements of each power set:

- i. $\mathcal{P}(\{0, 1\})$
- ii. $\mathcal{P}(\{0, 1, 2\})$
- iii. $\mathcal{P}(\{0\})$
- iv. $\mathcal{P}(\emptyset)$

State the order of each.

Problem 67. Let $A = \{\emptyset, \{\emptyset\}\}$. Decide, and justify, whether each is true or false.

- i. $\emptyset \in \mathcal{P}(A)$
- ii. $\emptyset \subseteq \mathcal{P}(A)$
- iii. $\{\emptyset\} \in \mathcal{P}(A)$
- iv. $\{\emptyset\} \subseteq \mathcal{P}(A)$
- v. $\{\{\emptyset\}\} \in \mathcal{P}(A)$
- vi. $\{\{\emptyset\}\} \subseteq \mathcal{P}(A)$

Problem 68. Prove or disprove: if $A \subseteq B$ then $\mathcal{P}(A) \subseteq \mathcal{P}(B)$.

Problem 69. Where A is a finite set, prove that $|\mathcal{P}(A)| = 2^{|A|}$.

2.2 Cartesian product

Definition 70. A sequence $\langle x_0, x_1, \dots, x_{n-1} \rangle$ is an ordered list. Its elements $x_0, x_1, \dots, x_{n-1}$ are terms. Its length $\text{lh}(\langle x_0, x_1, \dots, x_{n-1} \rangle)$ is the number of terms, n . Two sequences are equal if and only if they have the same length and the same terms, in the same order.

Problem 71. True or false?

- i. $\langle 1, 3, 5 \rangle = \langle 5, 3, 1 \rangle$
- ii. $\langle 2, 4, 6 \rangle = \langle 2, 4, 6, 4 \rangle$

Definition 72. For sets $A_0, A_1, \dots, A_{n-1}$, the collection of all sequences $\langle a_0, a_1, \dots, a_{n-1} \rangle$ where $a_0 \in A_0, a_1 \in A_1, \dots$, is the Cartesian product, denoted $A_0 \times A_1 \times \dots \times A_{n-1}$. If the sets are equal we write $A^n = A \times \dots \times A$.

A sequence of length two is often called an *ordered pair* and written with parentheses (x_0, x_1) (similarly we have ordered triples, four-tuples, etc.). Thus we may write $\mathbb{R}^2 = \{(x, y) \mid x, y \in \mathbb{R}\}$ for the Cartesian plane.

Problem 73. *Prove that $\mathbb{N}^2 \subseteq \mathbb{Z}^2$. Generalize.*

Problem 74. *(Algebra of Cartesian Product)*

- A. *Prove that $A \times B = \emptyset$ iff $A = \emptyset$ or $B = \emptyset$.*
- B. *Show that there are sets so that $A \times B \neq B \times A$. Under what circumstances are they equal?*
- C. *Show that this is false: $A \times B \subseteq \hat{A} \times \hat{B}$ if and only if $A \subseteq \hat{A}$ and $B \subseteq \hat{B}$. Patch it to make it true.*

Problem 75. *(Interaction of Cartesian product with other set operations)*

- A. *Prove that $(A \cup B) \times C = (A \times C) \cup (B \times C)$. What about intersection?*
- B. *Show that in general $(A \times B)^c$ does not equal $A^c \times B^c$.*

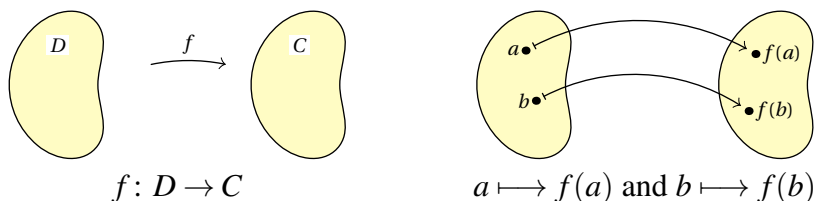
Chapter 3

Functions and relations

Definition 76. A function or map f from domain set D to codomain set C , written $f: D \rightarrow C$, is a triple consisting of the two sets along with a graph, a set of pairs $(d, c) \in D \times C$. The function must be well-defined: for each $d \in D$ there must be exactly one $c \in C$ such that (d, c) is an element of the graph. Functions are equal only if they have the same domain, codomain, and graph.

Thus, a function associates each element d from the domain, called an *argument* or *input*, with an element c from the codomain, called a *value* or *output*, subject to the condition that d determines c . We write $f(d) = c$ or $d \mapsto c$ and say that c is the *image* of d or that d *maps to* c .

A *bean diagram* pictures sets as blobs and either shows the entire function as a simple arrow, or else shows the function's action on individual elements with arrows that begin with a bar.



Problem 77. Decide if each is a function.

- i. $D = \{0, 1, 2\}$, $C = \{3, 4, 5\}$, $G = \{(0, 3), (1, 4), (2, 5)\}$
- ii. $D = \{0, 1, 2\}$, $C = \{3, 4, 5\}$, $G = \{(0, 3), (1, 4), (2, 3)\}$
- iii. $D = \{0, 1, 2\}$, $C = \{3, 4, 5\}$, $G = \{(0, 3), (1, 4)\}$
- iv. $D = \{0, 1, 2\}$, $C = \mathbb{N}$, $G = \{(0, 3), (1, 3), (2, 3)\}$
- v. $D = \mathbb{N}$, $C = \mathbb{N}$, $G = \{(0, 3), (1, 4), (2, 5)\}$

$$\text{vi. } D = \{0, 1, 2\}, C = \{3, 4, 5\}, G = \{(0, 3), (1, 4), (2, 4), (0, 5)\}$$

$$\text{vii. } D = \mathbb{N}, C = \mathbb{N}, G = \{(d, c) \in D \times C \mid c = d^2\}$$

Do not think that a function must have a formula. The final item in the prior exercise has a formula but for other items the graph G just has arbitrary pairings.

Problem 78. The hailstone function $h: \mathbb{N} \rightarrow \mathbb{N}$ is defined by cases.

$$h(n) = \begin{cases} n/2 & \text{-- if } n \text{ is even} \\ 3n + 1 & \text{-- otherwise} \end{cases}$$

i. Compute $h(n)$ for $n = 0, \dots, n = 9$.

ii. Iterate the function starting with input 6, that is, compute $h(6)$, then $h(h(6))$, etc., until the result is 1. How many steps does it take?

iii. How many steps does it take starting with $n = 11$?

The Collatz conjecture is that for every starting value greater than zero, iteration will eventually reach 1. No one knows if it is true.

We often blur the distinction between a function and its graph. For instance we may say, “a function is an input-output relationship” when technically it is the function’s graph that is the pairing. (The distinction is there only because the graph does not determine the codomain and so we must specify it separately. The graph does, however, determine the domain.)

In the edge case that the domain or codomain is empty, the only function has an empty graph.

Problem 79. Show that $\{(x, y) \in \mathbb{R}^2 \mid y^2 = x\}$ is not the graph of a function.

Problem 80. When D and C are finite sets, how many functions are there from D to C ?

Definition 81. The characteristic function of a set A is a map χ_A , whose domain is the universe, such that $\chi_A(x) = 1$ if $x \in A$ and $\chi_A(x) = 0$ if $x \notin A$.

A function may have multiple arguments; one example is the function $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ whose action is $\langle x, y \rangle \mapsto x^2 - 2y^2$. We write $f(x, y)$ rather than $f(\langle x, y \rangle)$. We say that this f is 2-ary and similarly there are 3-ary functions, etc. The number of arguments is the function’s *arity*.

Definition 82. The range of $f: D \rightarrow C$ is

$$\text{Ran}(f) = \{y \in C \mid \text{there is an } x \in D \text{ such that } f(x) = y\}$$

(it is also denoted $f(D)$).

Problem 83. For each item in Problem 77, if it is a function then find its range.

Definition 84. Let $f: D \rightarrow C$. The restriction of f to the subset $B \subseteq D$ is the function $f|_B: B \rightarrow C$ given by $f|_B(b) = f(b)$ for all $b \in B$ (we also say that f is an extension of $f|_B$). The image of the subset, denoted $\text{Im}(f)$ or $f(B)$, is the range of $f|_B$. In the other direction, the inverse image of the element $c \in C$ is the set $f^{-1}(c) = \{d \in D \mid f(d) = c\}$, and the inverse image of the set $A \subseteq C$ is $f^{-1}(A) = \{d \in D \mid f(d) \in A\}$.

Observe that in that definition $f^{-1}(c)$ is a set, not an element.

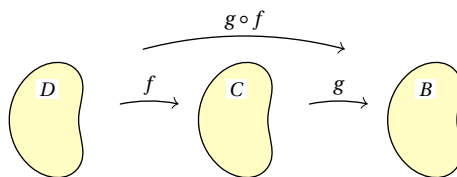
Problem 85. Where $f: \mathbb{R} - \{(2n+1) \cdot \pi/2 \mid n \in \mathbb{Z}\} \rightarrow \mathbb{R}$ is the function $f(x) = \tan(x)$,

- i. Find the image under f of the interval $[\pi/4 \dots \pi/2) = \{x \in \mathbb{R} \mid \pi/4 \leq x < \pi/2\}$.
- ii. Find the image of the single-element set $\{-\pi/3\}$.
- iii. Find the inverse image of the number 1.

Problem 86. Prove that $f^{-1}(A)$ is the union of the sets $f^{-1}(a)$ over all $a \in A$.

3.1 Composition

Definition 87. The composition of two functions $f: D \rightarrow C$ and $g: C \rightarrow B$ is $g \circ f: D \rightarrow B$ given by $g \circ f(d) = g(f(d))$.



Read $g \circ f$ aloud as “ g composed with f ” (or “ g circle f ” or “ g following f ”). Note that there is an awkwardness about the expression $g \circ f$: while you read the g first, the function that you apply first is f . Note also that the domain of g is the codomain of f . We are sometimes casual about this and don’t object as long as the domain of g includes the range of f .

Problem 88. Let $D = \{0, 1, 2\}$, $C = \{10, 11, 12, 13\}$, and $B = \{20, 21, 22\}$. Let $f: D \rightarrow C$ be given by $0 \mapsto 10$, $1 \mapsto 12$, $2 \mapsto 13$ and let $g: C \rightarrow B$ be $10 \mapsto 20$, $11 \mapsto 21$, $12 \mapsto 22$, and $13 \mapsto 20$.

- i. Compute $g \circ f$ on all arguments or show that the composition is not defined.
- ii. Compute $f \circ g$ on all arguments or show that it is not defined.
- iii. Find the range of f and g , as well as of $f \circ g$ and $g \circ f$ if they are defined.

Problem 89. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be $f(x) = x^2$ and let $g: \mathbb{R} \rightarrow \mathbb{R}$ be $g(x) = 3x + 1$. Find the domain, codomain, and a formula for $g \circ f$ and $f \circ g$.

Problem 90. Prove each.

- i. (Associativity) $h \circ (g \circ f) = (h \circ g) \circ f$
- ii. Function composition need not be commutative.

3.2 Inverse

The definition of function specifies that for every input element there is exactly one associated output. This is asymmetric because the definition puts no such condition on output elements.

Definition 91. A function $f: D \rightarrow C$ is one-to-one, or 1-1, or an injection, if for each value there is at most one associated argument, that is, if $f(d_0) = f(d_1)$ for $d_0, d_1 \in D$ implies that $d_0 = d_1$. The function is onto, or a surjection, if for each member of the codomain $c \in C$ there is at least one associated argument, at least one $d \in D$ such that $f(d) = c$. A function that is both one-to-one and onto, so that for every member of the codomain there is exactly one associated argument, is a correspondence or bijection.

Problem 92. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be $f(x) = 3x + 1$ and $g: \mathbb{R} \rightarrow \mathbb{R}$ be $g(x) = x^2 + 1$.

- A. Show that f is one-to-one and onto.
- B. Show that g is not one-to-one and not onto.

Problem 93. Let D and C be finite sets. Prove that if there is a correspondence $f: D \rightarrow C$ then the two have the same number of elements. Hint: for both you can do induction on $|C|$ or $|D|$.

- A. If f is one-to-one then $|C| \geq |D|$.
- B. If f is onto then $|C| \leq |D|$.

Problem 94. (*Pigeonhole Principle*) Show that if you have $n > 0$ -many pigeonholes and more than n -many papers then at least one hole gets at least two papers.

Problem 95. Prove.

- A. A composition of one-to-one functions is one-to-one.
- B. A composition of onto functions is onto. With the prior item this gives that a composition of correspondences is a correspondence.
- C. If $g \circ f$ is one-to-one then f is one-to-one.
- D. If $g \circ f$ is onto then g is onto.
- E. If $g \circ f$ is onto, is f onto? If it is one-to-one, is g one-to-one?

Definition 96. An identity function $\text{id}: D \rightarrow D$ has the action $\text{id}(d) = d$ for all $d \in D$.

Definition 97. Given $f: D \rightarrow C$, if $g \circ f$ is the identity function then g is a left inverse function of f , or what is the same thing, f is a right inverse of g . If g is both a left and right inverse of f then it is an inverse (or two-sided inverse) of f , denoted f^{-1} .

Problem 98. Show each.

- A. Let $g: \mathbb{R}^3 \rightarrow \mathbb{R}^2$ be the projection map given by $(x, y, z) \mapsto (x, y)$ and let $f: \mathbb{R}^2 \rightarrow \mathbb{R}^3$ be the injection map given by $(x, y) \mapsto (x, y, 0)$. Then g is a left inverse of f but not a right inverse.
- B. The function $f: \mathbb{Z} \rightarrow \mathbb{Z}$ given by $f(n) = n^2$ has no left inverse.
- C. Where $D = \{0, 1, 2, 3\}$ and $C = \{10, 11\}$, the function $f: D \rightarrow C$ given by $0 \mapsto 10, 1 \mapsto 11, 2 \mapsto 10, 3 \mapsto 11$ has more than one right inverse.

Problem 99. i. Where $f: \mathbb{Z} \rightarrow \mathbb{Z}$ is $f(a) = a + 3$ and $g: \mathbb{Z} \rightarrow \mathbb{Z}$ is $g(a) = a - 3$, show that g is inverse to f .

ii. Where $h: \mathbb{Z} \rightarrow \mathbb{Z}$ is the function that returns $n + 1$ if n is even and returns $n - 1$ if n is odd, find a function inverse to h .

iii. If $s: \mathbb{R}^+ \rightarrow \mathbb{R}^+$ is $s(x) = x^2$, find its inverse.

Problem 100. Let $D = \{0, 1, 2\}$ and $C = \{10, 11, 12\}$. Also let $f, g: D \rightarrow C$ be $f(0) = 10, f(1) = 11, f(2) = 12$, and $g(0) = 10, g(1) = 10, g(2) = 12$.

- i. Verify that f is a correspondence.

- ii. *Construct an inverse for f .*
- iii. *Verify that g is not a correspondence.*
- iv. *Show that g has no inverse.*

In Definition 84, we wrote $f^{-1}(c)$ for the set $\{d \in D \mid f(d) = c\}$. This earlier notation is standard but it conflicts with what we just saw in Definition 97. For instance, if $g: \mathbb{R} \rightarrow \mathbb{R}$ is $g(x) = 2x$ then $g^{-1}(8)$ could mean two things: the earlier definition has $g^{-1}(8) = \{4\}$ while the above definition has $g^{-1}(8) = 4$. The difference between the set containing one element and that one element doesn't cause much trouble. However when the function is not one-to-one then we must be careful. For $\hat{g}: \mathbb{R} \rightarrow \mathbb{R}$ given by $\hat{g}(x) = x^2$, the earlier definition has $\hat{g}^{-1}(9) = \{3, -3\}$. But because $\hat{g}$ is not one-to-one, it is not invertible, and so $\hat{g}^{-1}$ in the sense of the definition above doesn't even exist.

In short, if a function has an inverse then the distinction between the two definitions is minor. But if it has no inverse then, while the earlier definition still applies, the later definition does not. Put another way, the use of the symbol f^{-1} does not imply that the function has an inverse.

Problem 101. *Prove.*

- A. *A function has an inverse if and only if that function is a correspondence.*
- B. *If a function has an inverse then that inverse is unique.*
- C. *The inverse of a correspondence is a correspondence.*
- D. *If f and g are each invertible then so is $g \circ f$, and $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$.*

3.3 Relations

By definition, a function's graph is a set of pairs $\langle \text{argument}, \text{value} \rangle$ subject to the condition that the argument determines the value. We now generalize, by dropping that condition.

Definition 102. *A relation on sets $A_0, \dots, A_{n-1}$ is a subset of the Cartesian product, $R \subseteq A_0 \times \dots \times A_{n-1}$. If all of the sets are the same $A_0 = A_1 = \dots = A$ then we say it is a relation on A . If $n = 2$ then it is a binary relation on A . In this case, where $\langle a_0, a_1 \rangle \in R$ we say that a_0 is R -related to a_1 , sometimes written $a_0 R a_1$. For larger n we say R is n -ary, and call n the arity of the relation.*

Problem 103. List five elements of each relation:

- i. $\{\langle x, y \rangle \in \mathbb{N}^2 \mid x \text{ and } y \text{ have the same parity}\}$
- ii. less-than, $<$, as a binary relation on $\mathbb{N}$
- iii. $\{\langle x, y, z \rangle \in \mathbb{N}^3 \mid x^2 + y^2 = z^2\}$
- iv. $E = \{(x, y) \in A \times \mathcal{P}(A) \mid x \in y\}$ where $A = \{0, 1, 2\}$

Problem 104. Verify that for any function $f: D \rightarrow C$, the set $R_f = \{(x, y) \in D^2 \mid f(x) = f(y)\}$ is a binary relation. List six elements of R_f where f is the real function $f(x) = x^2$.

Definition 105. Let R be a binary relation on a set X . It is reflexive if $\langle x, x \rangle \in R$ for all $x \in X$. It is symmetric if $\langle x, y \rangle \in R$ implies that $\langle y, x \rangle \in R$ for all $x, y \in X$. And it is transitive if $\langle x, y \rangle \in R$ and $\langle y, z \rangle \in R$ implies that $\langle x, z \rangle \in R$ for $x, y, z \in X$. A relation that is all three is an equivalence.

Problem 106. For each, prove or disprove that it is reflexive, it is symmetric, and it is transitive.

- A. The “goes into” relation, $G = \{\langle d, m \rangle \in \mathbb{Z}^2 \mid d \mid m\}$.
- B. For any set A , the diagonal relation, $\Delta_A = \{\langle a, a \rangle \mid a \in A\}$.
- C. The relation on the reals of “at least two greater,” $T = \{\langle x, y \rangle \in \mathbb{R}^2 \mid x - y \geq 2\}$.

Problem 107. Fix $m \in \mathbb{Z}^+$. Show that $M = \{\langle a, b \rangle \in \mathbb{Z}^2 \mid a \equiv b \pmod{m}\}$ is an equivalence.

Problem 108. Let $\mathcal{L}$ be the set of lines in the Euclidean plane and consider the relation $R = \{\langle \ell_0, \ell_1 \rangle \in \mathcal{L}^2 \mid \text{the two are parallel or equal}\}$.

- i. List five elements of R .
- ii. Where ℓ is a vertical line, list five elements of $\mathcal{L}$ that are related to ℓ .
- iii. Show that R is an equivalence.

Problem 109. There are sixteen binary relations on $A = \{0, 1\}$. Produce them and characterize each as reflexive or not, symmetric or not, and transitive or not.

Problem 110. Binary relations can be reflexive or not, symmetric or not, and transitive or not, so there are eight possible combinations.

- A. Four of these are not reflexive. For each, give an example relation on $A = \{0, 1, 2\}$.
- B. Give examples of relations on $A = \{0, 1, 2\}$ for the other four combinations.

Problem 111. Let two elements $\langle n_0, d_0 \rangle$ and $\langle n_1, d_1 \rangle$ of $\mathbb{Z} \times \mathbb{Z}^+$ be related if $n_0 d_1 = d_0 n_1$. List five elements of this relation. Prove that it is an equivalence.

Definition 112. If R is an equivalence relation on X then we sometimes write $x \equiv y \pmod{R}$, or $x \sim y$, in place of $\langle x, y \rangle \in R$. The equivalence class of $x \in X$ is the set $\llbracket x \rrbracket = \{y \in X \mid y \equiv x \pmod{R}\}$.

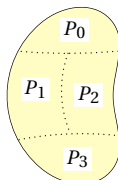
A common stumbling block is that $\llbracket x_0 \rrbracket = \llbracket x_1 \rrbracket$ does not imply that $x_0 = x_1$. An example is the relation of leaving the same remainder when divided by ten, $R = \{\langle x, y \rangle \in \mathbb{N}^2 \mid x \bmod 10 = y \bmod 10\}$. The set of numbers that leave a remainder of 1 is $\{1, 11, 21, 31, \dots\}$ and we could identify this set as the equivalence class of 1, or the equivalence class of 11, etc., $\llbracket 1 \rrbracket = \llbracket 11 \rrbracket = \llbracket 21 \rrbracket = \dots$.

Problem 113. Verify that each relation is an equivalence. Exhibit the equivalence classes.

- A. Two numbers $x_0, x_1 \in \mathbb{N}$ are related if they have the same parity.
- B. Two integers $m, n \in \mathbb{Z}$ are related if they leave the same remainder on division by 3, that is, if $m \equiv n \pmod{3}$.
- C. Two real numbers $r_0, r_1 \in \mathbb{R}$ are related if $r_0 - r_1 \in \mathbb{Z}$.

Problem 114. Let R be an equivalence on X . Prove that the following are equivalent statements for $x_0, x_1 \in X$: (i) $x_0 \equiv x_1 \pmod{R}$, (ii) $\llbracket x_0 \rrbracket = \llbracket x_1 \rrbracket$, and (iii) $\llbracket x_0 \rrbracket \cap \llbracket x_1 \rrbracket \neq \emptyset$.

Definition 115. A partition $\mathcal{P}$ of a set X is a collection of parts $P_i \subseteq X$ that are nonempty and such that every element $x \in X$ is in exactly one of the P_i 's. That is, each $P_i \in \mathcal{P}$ is nonempty, and $\mathcal{P}$ covers X (the union of all the P_i 's is X), and the parts are pairwise disjoint (if $P_i \cap P_j \neq \emptyset$ then $P_i = P_j$).



A set partitioned into four subset parts $\mathcal{P} = \{P_0, P_1, P_2, P_3\}$

Problem 116. Verify that $\mathcal{P}$ is a partition of X . How many parts does it have?

- A. $X = \mathbb{N}$, $\mathcal{P} = \{P_0, P_1\}$, where P_0 is the set of even numbers and P_1 is the set of odd numbers.
- B. $X = \mathbb{Z}$, $\mathcal{P} = \{P_n \mid n \in \mathbb{Z}\}$ where $P_n = \{i \in \mathbb{Z} \mid i \equiv n \pmod{3}\}$
- C. $X = \mathbb{R}$, $\mathcal{P} = \{P_x \mid x \in \mathbb{R}\}$ where $P_x = \{y \in \mathbb{R} \mid x - y \in \mathbb{Z}\}$

Problem 117. Prove.

- A. Where R is an equivalence on the set X , the collection of equivalence classes $\{\llbracket x \rrbracket \mid x \in X\}$ forms a partition of X . This is the partition induced by the relation.
- B. Where $\mathcal{P}$ is a partition of X , the relation

$$R = \{\langle x, y \rangle \in X^2 \mid x \text{ and } y \text{ are in the same part}\}$$

is an equivalence. This is the equivalence relation that arises from the partition.

Problem 118. Suppose that $f: D \rightarrow C$.

- A. Show that the relation $R_f = \{(d_0, d_1) \in D^2 \mid f(d_0) = f(d_1)\}$ is an equivalence on D .
- B. Prove that the set of inverse images $\mathcal{P} = \{f^{-1}(c) \mid c \in \text{Ran}(f)\}$ partitions the domain.
- C. Using that partition, consider $\hat{f}: \mathcal{P} \rightarrow \text{Ran}(f)$ defined by: $\hat{f}(P) = f(d)$ for any $d \in P$. Show that $\hat{f}$ is well-defined and one-to-one. Remark: any function can be modified to be onto by setting its codomain to be its range. Here we also change the domain to get one-to-one.

Definition 119. A binary relation R is antisymmetric if $\langle x, y \rangle \in R$ and $\langle y, x \rangle \in R$ implies that $x = y$. A binary relation is a partial ordering if it is reflexive, antisymmetric, and transitive.

Problem 120. Verify each.

- A. The usual less than or equal to relation, $\leq$, on the real numbers is a partial order.
- B. The relation “divides” on $\mathbb{N}$ is a partial order.
- C. For any set A the relation $\subseteq$ on $\mathcal{P}(A)$ is a partial order.

Problem 121. Can a relation be both symmetric and antisymmetric?

Chapter 4

Infinity

Recall Problem 93, that if two finite sets correspond then they have the same number of elements.

Definition 122. Two sets have the same cardinality or are equinumerous if there is a correspondence from one to the other. We write $A \sim B$.

Problem 123. For each, prove the two have the same cardinality.

- i. $\mathbb{N}$ and $\mathcal{E} = \{2k \mid k \in \mathbb{N}\}$,
- ii. $(-\pi/2 \dots \pi/2)$ and $\mathbb{R}$,
- iii. $(0 \dots 1)$ and $\mathbb{R}$

Problem 124. Prove that the relation $\sim$ is an equivalence.

Definition 125. A set is finite if it has n elements for some $n \in \mathbb{N}$, that is, if it has the same cardinality as some initial segment $\{i \in \mathbb{N} \mid i < n\} = \{0, 1, \dots, n-1\}$ of the natural numbers. Otherwise the set is infinite. A set is denumerable if it has the same cardinality as $\mathbb{N}$. A set is countable if it is either finite or denumerable.

Problem 126. Prove each.

- A. The set of integers is countable.
- B. The set $\mathbb{N} \times \mathbb{N}$ is countable.

Problem 127. Prove that the following are equivalent for a set A : (i) A is countable, (ii) A is empty or there is an onto function from $\mathbb{N}$ to A , (iii) there is a one-to-one function from A to $\mathbb{N}$.

Problem 128. Prove that the set of rational numbers is countable.

Problem 129. *Prove that each of these infinite sets is not countable.*

- A. $\mathcal{P}(\mathbb{N})$ *Hint: suppose that $f: \mathcal{P}(\mathbb{N}) \rightarrow \mathbb{N}$ and consider $\{n \in \mathbb{N} \mid n \notin f(n)\}$.*
- B. $\mathbb{R}$ *Hint: find a one-to-one map from $\mathcal{P}(\mathbb{N})$ to $\mathbb{R}$.*